INCH-POUND
MIL-DTL-24643/29F
27 November 2023
SUPERSEDING
MIL-DTL-24643/29E
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 300 VOLTS, TYPE LSMU (INCLUDING VARIATION LSMUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LSMU and LSMUS (double overall shield).

Construction (Non-Watertight)

First - Copper conductor, tin-coated, ASTM B286, size 20 through size 7 strands.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Thermoset insulation. Minimum average wall thickness of 0.013 inch. Standard identification code applied by method 1 on white-colored insulation.

Fourth - The specified number of singles (see table I) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly.

Fifth - An optional binder.

Construction for Variant Type LSMU

Sixth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.030 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Construction for Variant Type LSMUS

Sixth - Braided shield of tin-coated copper.

Seventh - Spirally wrapped, 0.002-inch-thick polyethylene terephthalate (polyester film) separator tape with 25 percent minimum overlap.

Eighth - Braided shield of tin-coated copper.

Ninth - An optional binder.

Distribution Statement A. Approved for public release: distribution is unlimited.

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Tenth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.030 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details (shielded and unshielded).

Military part no. M24643/29	Type and size	Number of conductors	Overall diameter (inch)	
			min.	max.
-01UN	LSMU-14	14	0.365	0.400
-01UD	LSMUS-14	14	0.425	0.460

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	10.93
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,000
Conductor to overall shield (type LSMUS only)	1,000
Overall shield to overall shield (type LSMUS only)	200
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	2,000
Conductor to overall shield (type LSMUS only)	2,000
Overall shield to overall shield (type LSMUS only)	100
Conductor continuity	No failure
Shield continuity (type LSMUS only)	No failure
Jacket flaws (type LSMUS only)	No failure

Group A:

Visual and dimensional	No failure

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Group B:	
Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Surface transfer impedance (type LSMUS only)	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60
Group D:	
Flame propagation (cable)	No failure

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Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSMU-14 and LSMUS-14	2,500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-045)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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